

Directed Graphs

DIRECTED GRAPHS

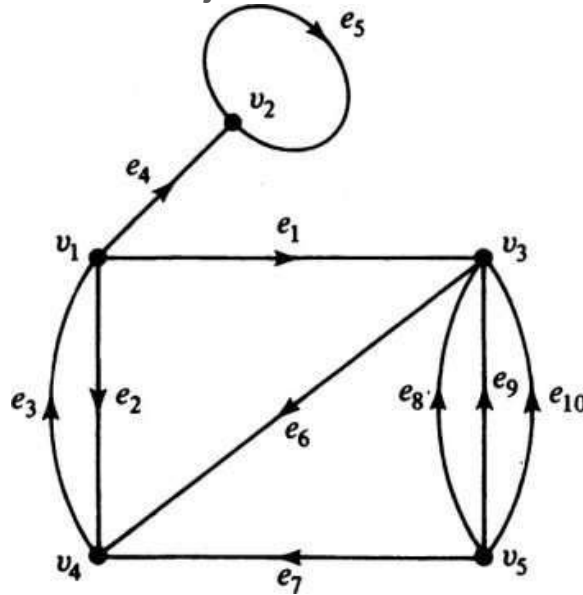
The graphs we have seen so far have been undirected graphs. No direction was assigned to the edges in a graph. An edge e_k between vertices v_i and v_j could be considered as going from vertex v_i to vertex v_j or from v_j to v_i .

In this chapter we shall consider directed graphs. In a directed graph edges have specific directions.

Most of the concepts and terminology of undirected graphs are also applicable to directed graphs. For example, the planarity of a graph does not depend on whether the graph is directed or undirected, and therefore Chapter 5 is applicable to both directed and undirected graphs.

WHAT IS A DIRECTED GRAPH?

A directed graph (or a digraph for short) G consists of a set of vertices $V = \{v_1, v_2, \dots\}$, a set of edges $E = \{e_1, e_2, \dots\}$, and a mapping Ψ that maps every edge onto some **ordered pair of vertices** (v_i, v_j) .



WHAT IS A DIRECTED GRAPH?

In a digraph an edge is not only incident on a vertex, but is also incident out of a vertex and incident into a vertex. The vertex v_i , which edge e_k is incident out of, is called the initial vertex of e_k . The vertex v_j , which e_k is incident into, is called the terminal vertex of e_k .

An edge for which the initial and terminal vertices are the same forms a self-loop, such as e_5 in the graph shown earlier.

WHAT IS A DIRECTED GRAPH?

The number of edges incident out of a vertex v_i is called the out-degree and is written $d^+(v_i)$.

The number of edges incident into v_i is called the in-degree and is written as $d^-(v_i)$.

For example, in the previous graph

$$d^+(v_1) = 3, d^-(v_1) = 1,$$

$$d^+(v_2) = 1, d^-(v_2) = 2,$$

$$d^+(v_5) = 4, d^-(v_5) = 0$$

WHAT IS A DIRECTED GRAPH?

It is not difficult to prove that in any digraph G the sum of all in-degrees is equal to the sum of all out-degrees, each sum being equal to the number of edges in G ; that is,

$$\sum_{i=1}^n d^+(v_i) = \sum_{i=1}^n d^-(v_i).$$

An isolated vertex is a vertex in which the in-degree and the out-degree are both equal to zero. A vertex v in a digraph is called pendant if it is of degree one, that is, if

$$d^+(v) + d^-(v) = 1$$

WHAT IS A DIRECTED GRAPH?

Two directed edges are said to be parallel if they are mapped onto the same ordered pair of vertices.

That is, in addition to being parallel in the sense of undirected edges, parallel directed edges must also agree in the direction of their arrows. In the graph shown earlier, edges e_8 , e_9 , and e_{10} are parallel, whereas edges e_2 and e_3 are not.

WHAT IS A DIRECTED GRAPH?

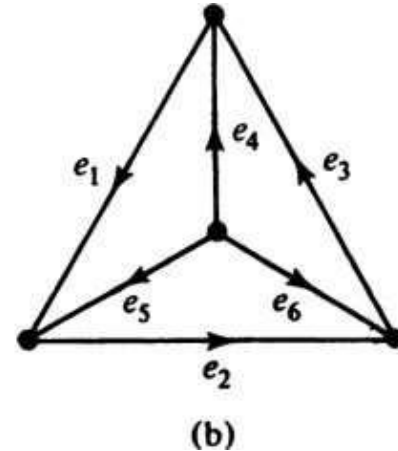
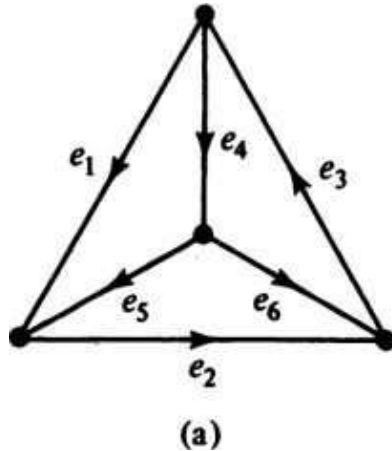
Since many properties of directed graphs are the same as those of undirected ones, it is often convenient to disregard the orientations of edges in a digraph. Such an undirected graph obtained from a directed graph G will be called the **undirected graph corresponding to G** .

Now we answer the question: **When are two digraphs considered to be the same or isomorphic?**

Isomorphic Digraphs

For two digraphs to be isomorphic not only must their corresponding undirected graphs be isomorphic, but the directions of the corresponding edges must also agree.

For example, The figure below shows two digraphs that are not isomorphic, although they are orientations of the same undirected graph.



TYPES OF DIGRAPHS

Simple Digraphs: A digraph that has no self-loop or parallel edges is called a simple digraph.

Asymmetric Digraphs: Digraphs that have at most one directed edge between a pair of vertices, but are allowed to have self-loops, are called asymmetric or antisymmetric.

Symmetric Digraphs: Digraphs in which for every edge (a, b) (i.e., from vertex a to b) there is also an edge (b, a) .

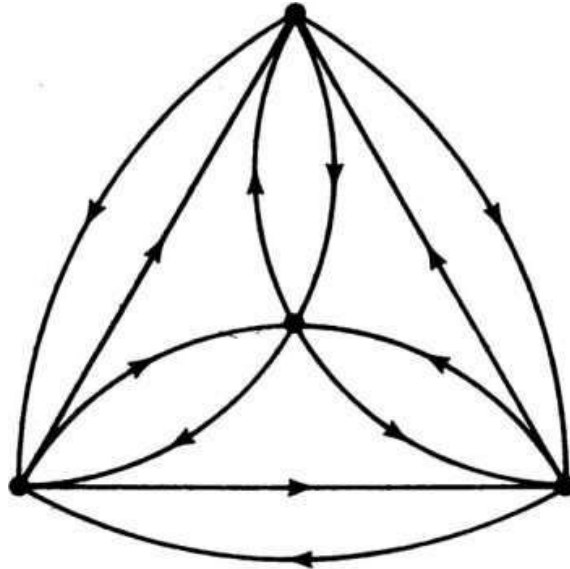
TYPES OF DIGRAPHS

Complete Digraphs: A complete undirected graph was defined as a simple graph in which every vertex is joined to every other vertex exactly by one edge.

For digraphs we have two types of complete graphs. A **complete symmetric digraph** is a simple digraph in which there is exactly one edge directed from every vertex to every other vertex, and a **complete asymmetric digraph** is an asymmetric digraph in which there is exactly one edge between every pair of vertices.

TYPES OF DIGRAPHS

A complete asymmetric digraph of n vertices contains $n(n - 1)/2$ edges, but a complete symmetric digraph of n vertices contains $n(n - 1)$ edges. A complete asymmetric digraph is also called a **tournament** or a **complete tournament**.



TYPES OF DIGRAPHS

Balanced digraph: A digraph is said to be balanced if for every vertex v_i the in-degree equals the out-degree; that is,

$$d^+(v_i) = d^-(v_i)$$

Regular digraph: A balanced digraph is said to be regular if every vertex has the same in-degree and out-degree as every other vertex; that is

$$d^+(v_i) = d^-(v_i) = k$$

DIGRAPHS AND BINARY RELATIONS

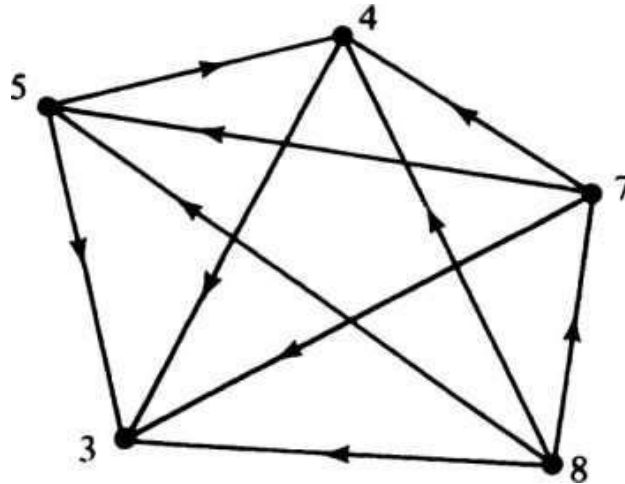
In a set of objects, X , where $X = \{x_1, x_2, \dots\}$, a binary relation R between pairs (x_i, x_j) may exist. In which case, we write $x_i R x_j$ and say that x_i has relation R to x_j .

Relation R may for instance be “is parallel to,” “is orthogonal to,” or “is congruent to” in geometry. It may be “is greater than,” “is a factor of,” “is equal to,” and so on, in the case when X consists of numbers. On the other hand, if the set X is composed of people, the relation R may be “is son of,” “is spouse of,” “is friend of,” and so forth.

Each of these relations is defined only on pairs of numbers of the set, and this is why the name binary relation.

DIGRAPHS AND BINARY RELATIONS

A digraph is the most natural way of representing a binary relation on a set X . Each $x_i \in X$ is represented by a vertex x_i . If x_i has the specified relation R to x_j , a directed edge is drawn from vertex x_i to x_j , for every pair (x_i, x_j) . For example, the digraph below represents the relation “is greater than” on a set consisting of five numbers $\{3, 4, 7, 5, 8\}$.



DIGRAPHS AND BINARY RELATIONS

Reflexive Relation: For some relation R it may happen that every element is in relation R to itself. For example, a number is always equal to itself, or a line is always parallel to itself.

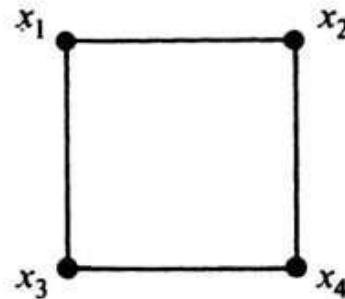
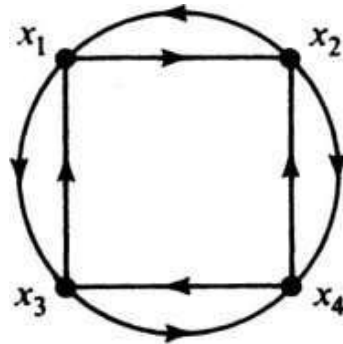
Such a relation R on set X that satisfies $x_i R x_i$ for every $x_i \in X$ is called a reflexive relation. The digraph of a reflexive relation will have a self-loop at every vertex.

Such a digraph representing a reflexive binary relation on its vertex set may be called a reflexive digraph. A digraph in which no vertex has a self-loop is called an irreflexive digraph.

DIGRAPHS AND BINARY RELATIONS

Symmetric Relation: For some relation R it may happen that for all x_i and x_j if $x_i R x_j$ holds, then $x_j R x_i$ also holds. Such a relation is called a symmetric relation. “Is spouse of” is a symmetric but irreflexive relation. “Is equal to” is both symmetric and reflexive.

The digraph of a symmetric relation is a symmetric digraph because for every directed edge from vertex x_i to x_j there is a directed edge from x_j to x_i .



DIGRAPHS AND BINARY RELATIONS

Transitive Relation: A relation R is said to be transitive if for any three elements x_i , x_j , and x_k in the set,

$$x_i R x_j \text{ and } x_j R x_k$$

always imply

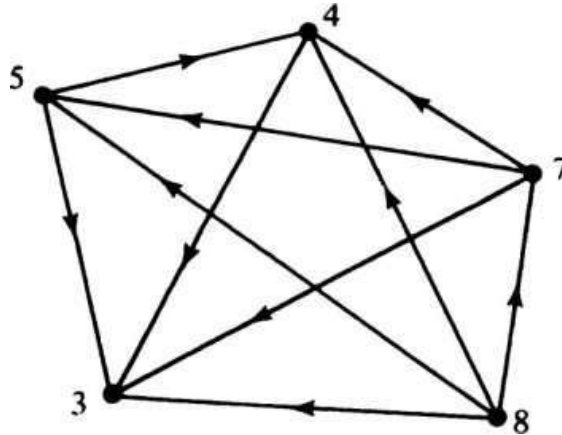
$$x_i R x_k$$

The binary relation “is greater than,” for example, is a transitive relation. If $x_i > x_j$ and $x_j > x_k$, clearly $x_i > x_k$.

“Is descendent of” is another example of a transitive relation.

DIGRAPHS AND BINARY RELATIONS

The digraph of a transitive (but irreflexive and asymmetric) binary relation is shown below.

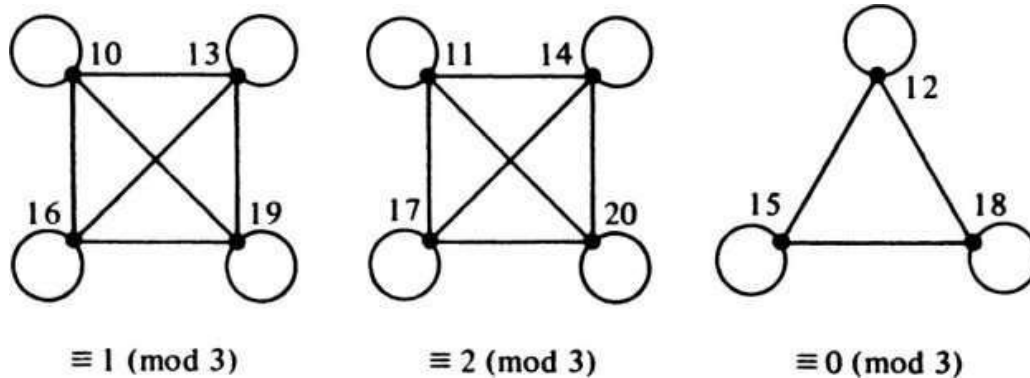


Note the triangular subgraphs. A digraph representing a transitive relation (on its vertex set) is called a transitive directed graph.

DIGRAPHS AND BINARY RELATIONS

Equivalence Relation: A binary relation is called an equivalence relation if it is simultaneously reflexive, symmetric, and transitive. Some examples of equivalence relations are “is parallel to,” “is equal to,” “is congruent to,” “is equal to modulo m ,” and “is isomorphic to.”

What does an equivalence graph look like? Here we see an equivalence relation “is congruent to modulo 3” defined on the set of 11 integers, 10 through 20.



DIGRAPHS AND BINARY RELATIONS

Relation Matrices: A binary relation R on a set can also be represented by a matrix, called a relation matrix. It is a $(0, 1)$, n by n matrix, where n is the number of elements in the set. The i, j th entry in the matrix is 1 if $x_i R x_j$ is true, and is 0, otherwise. For example, the relation matrix of the relation “is greater than” on the set of integers $\{3, 4, 7, 5, 8\}$ is

$$\begin{array}{c} \\ 3 \\ 4 \\ 7 \\ 5 \\ 8 \end{array} \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 \end{bmatrix}.$$

DIRECTED WALKS AND PATHS

There are actually two kinds of walks (and paths and circuits) in case of a digraph.

A **directed walk** from a vertex v_i to v_j is an alternating sequence of vertices and edges, beginning with v_i and ending with v_j , such that each edge is oriented from the vertex preceding it to the vertex following it.

Of course, no edge in a directed walk appears more than once, but a vertex may appear more than once, just as in the case of undirected graphs. If the walk contains no repetitions of vertices as well, then it is called a **directed path**.

A **semiwalk** in a directed graph is a walk in the corresponding undirected graph, but is not a directed walk.

CONNECTEDNESS OF DIGRAPHS

In a digraph there are two different types of paths. Consequently, we have two different types of connectedness in digraphs.

A digraph G is said to be **strongly connected** if there is at least one directed path from every vertex to every other vertex.

A digraph G is said to be **weakly connected** if its corresponding undirected graph is connected but G is not strongly connected.

CONNECTEDNESS OF DIGRAPHS

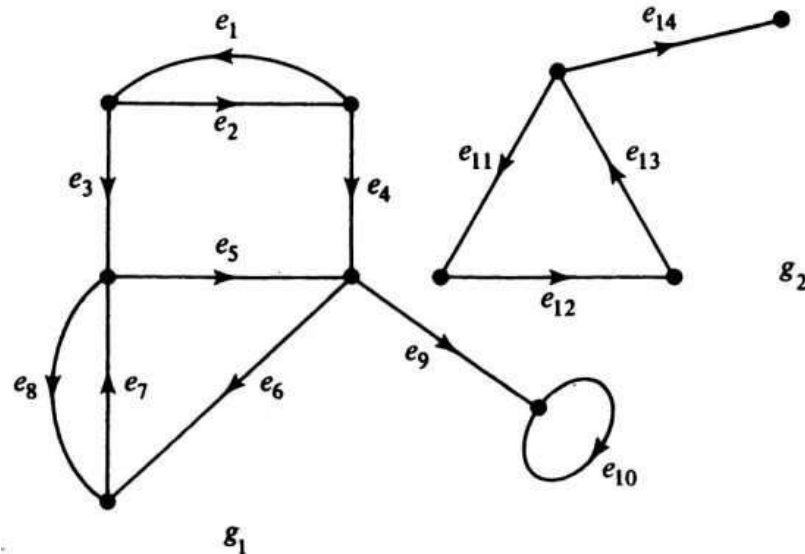
Since there are two types of connectedness in a digraph, we can define two types of components also.

Each maximal connected (weakly or strongly) subgraph of a digraph G will still be called a **component** of G .

But within each component of G the maximal strongly connected subgraphs will be called the **fragments** (or strongly connected fragments) of G .

CONNECTEDNESS OF DIGRAPHS

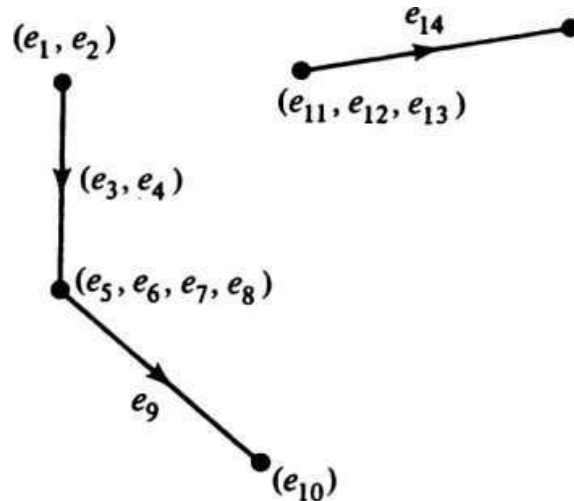
For example, the digraph shown here consists of two components. The component g_1 contains three fragments $\{e_1, e_2\}$, $\{e_5, e_6, e_7, e_8\}$, and $\{e_{10}\}$. Observe that e_3, e_4 , and e_9 do not appear in any fragment of g_1 .



Condensation

The condensation G_c of a digraph G is a digraph in which each strongly connected fragment is replaced by a vertex, and all directed edges from one strongly connected component to another are replaced by a single directed edge.

The condensation of the digraph shown earlier is as follows:



Condensation

Two observations can be made from the definition of condensation:

1. The condensation of a strongly connected digraph is simply a vertex.
2. The condensation of a digraph has no directed circuit.

Accessibility: In a digraph a vertex b is said to be accessible (or reachable) from vertex a if there is a directed path from a to b . Clearly, a digraph G is strongly connected if and only if every vertex in G is accessible from every other.

EULER DIGRAPHS

When is a digraph an Euler digraph?

Clearly, the digraph must be connected, with the possible exception of isolated vertices; otherwise, every edge cannot be traversed in one walk. In fact, an Euler digraph must be strongly connected, although every strongly connected digraph need not be an Euler digraph.

Thus, strongly connectedness is a necessary, but not sufficient condition for having an Euler digraph.

THEOREM 9-1: *“A digraph G is an Euler digraph if and only if G is connected and is balanced [i.e., $d^-(v) = d^+(v)$ for every vertex v in G].”*

Teleprinter's Problem

Let us now consider an application of the Euler digraph for solving an important problem in communication theory, which is often referred to as the teleprinter's problem.

The problem is stated as: **“How long is a longest circular (or cyclic) sequence of 1's and 0's such that no subsequence of r bits appears more than once in the sequence? Construct one such longest sequence.”**

Teleprinter's Problem

We start with the observation that since there are 2^r distinct r -tuples formed from 0 and 1, the sequence can be no longer than 2^r bits long.

Using Theorem 9-1, we shall construct a circular sequence 2^r bits long with the required property that no subsequence of r bits be repeated.

Construct a digraph G whose vertices are all $(r - 1)$ -tuples of 0's and 1's. Clearly, there are 2^{r-1} vertices in G . Let a typical vertex be

$$\alpha_1 \alpha_2 \dots \alpha_{r-1}, \text{ where } \alpha_i = 0 \text{ or } 1$$

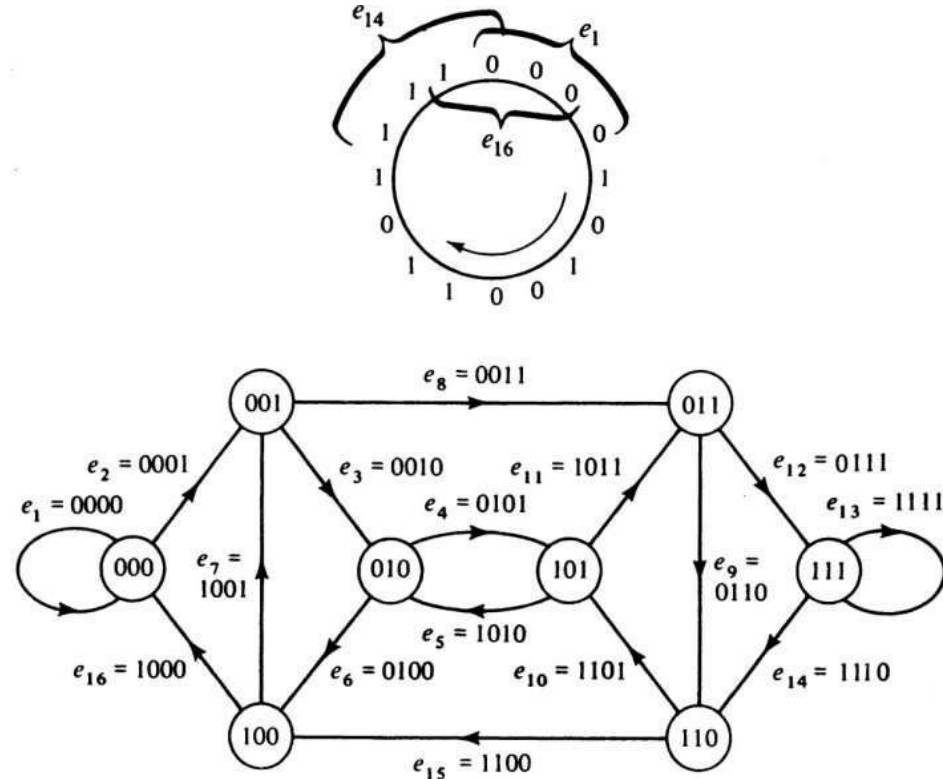
Draw an edge directed from this vertex to each of two vertices $(\alpha_2 \alpha_3 \dots \alpha_{r-1} 0)$ and $(\alpha_2 \alpha_3 \dots \alpha_{r-1} 1)$.

Teleprinter's Problem

In this way, draw two such edges directed from each of the 2^{r-1} vertices.
(A self-loop will result in each of the two cases when $\alpha_1 = \alpha_2 = \dots = \alpha_{r-1} = 0$ or 1 .)

The resulting digraph is an Euler digraph because for each vertex the in-degree equals the out-degree (each being equal to two).

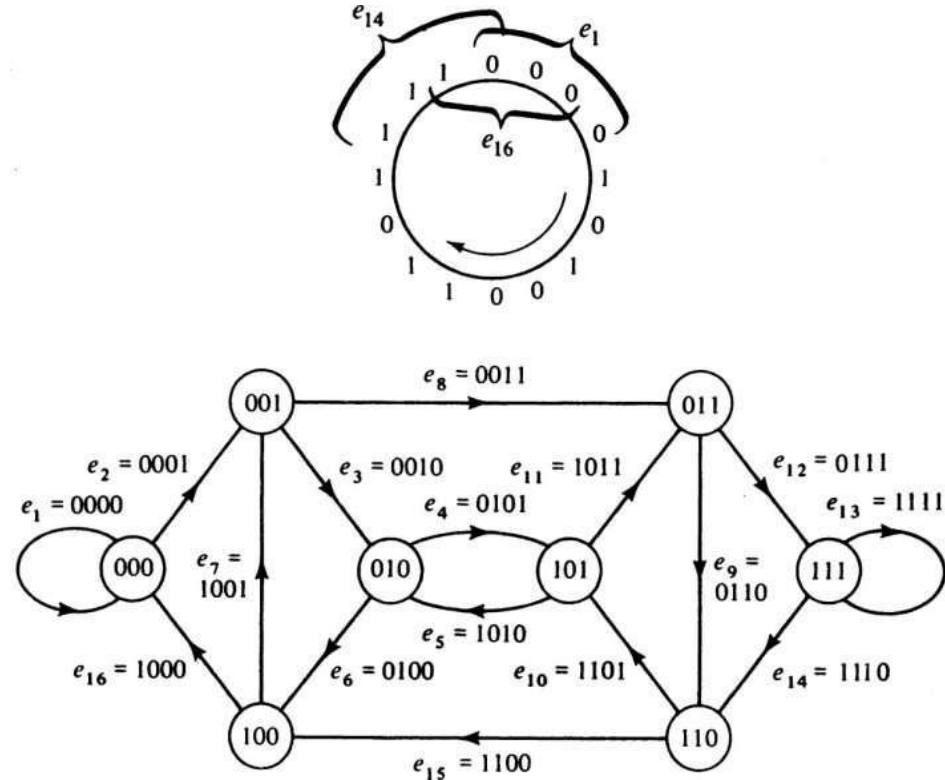
A directed Euler line in G consists of the 2^r edges, each with a distinct r -bit label.



Teleprinter's Problem

The figure given here illustrates the procedure of obtaining such a maximum-length sequence for $r = 4$.

One such sequence is
0000101001101111 corresponding to
the walk $e_1 e_2 e_3 e_4 e_5 e_6 e_7 e_8 e_9 e_{10} e_{11}$
 $e_{12} e_{13} e_{14} e_{15} e_{16}$.



Teleprinter's Problem

It is not difficult to see that the alphabet size need not be 2. It could be any number m .

In that case, the maximum-length sequence is m^r symbols long. The in-degree and out-degree of each vertex in the corresponding Euler digraph equals m , rather than two.

Another interesting task here is finding the number of **distinct directed Euler lines** in a given Euler digraph. In fact, the digraph shown earlier has 16 distinct Euler lines. (Note that rotations of the same sequence of edges are not considered distinct.)

TREES WITH DIRECTED EDGES

The number of distinct Euler lines in a balanced, connected digraph G can be obtained by counting certain types of spanning trees in G , which will be covered in this section.

We will start with defining trees containing directed edges. In addition to being trees in the undirected sense, trees in digraphs have additional properties and variations resulting from the relative orientations of the edges.

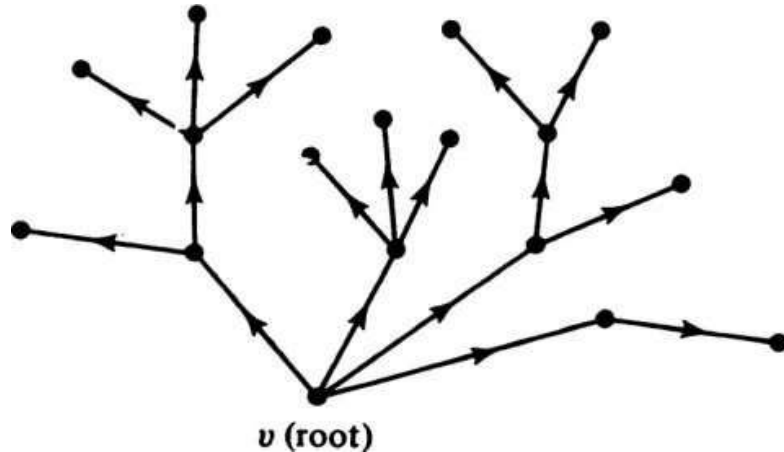
One particularly useful type of rooted tree with directed edges is called an **arborescence**.

TREES WITH DIRECTED EDGES

Arborescence: A digraph G is said to be an arborescence if

1. G contains no circuit – neither directed nor semicircuit.
2. In G there is precisely one vertex v of zero in-degree.

This vertex v of zero in-degree is called the **root** of the arborescence.



THEOREM 9-2

“An arborescence is a tree in which every vertex other than the root has an in-degree of exactly one.”

Proof: According to condition 1, an arborescence with n vertices can have at most $n - 1$ edges. Therefore, the sum of in-degrees of all vertices in G

$$d^-(v_1) + d^-(v_2) + \cdots + d^-(v_n) \leq n - 1$$

By condition 2, of the n terms on the left-hand side of this equation, only one is zero, and others must all be positive integers. Therefore, they must all be 1's.

Now, since there are exactly $n - 1$ vertices of in-degree one and one vertex of in-degree zero, digraph G has exactly $n - 1$ edges. Since G is also circuitless, it must be connected, and hence a tree.

Arborescence

An arborescence is in a sense a tree directed out of the root. Therefore, an arborescence is sometimes referred to as an **out-tree**.

Reversing the direction of every edge in an arborescence will produce what may be called an **in-tree**.

THEOREM 9-3: *“In an arborescence there is a directed path from the root R to every other vertex. Conversely, a circuitless digraph G is an arborescence if there is a vertex v in G such that every other vertex is accessible from v , and v is not accessible from any other vertex.”*

Application of Arborescence - Polish Notation

Consider the arithmetic expression

$$a + b - \frac{c \cdot d}{g^x - f}. \quad (9-1)$$

In a procedural language (such as FORTRAN or ALGOL) this expression might be written as

$$a + b - c \cdot d \div (g \uparrow x - f), \quad (9-2)$$

In evaluating this expression the computer must perform the arithmetic operations in a certain order; otherwise, it will produce a wrong result.

To evaluate such an expression, the machine will have to scan the expression back and forth to find the sequence of operations to be performed. To avoid scanning back and forth, the computer makes a preliminary translation of expressions into the Polish notation.

Application of Arborescence - Polish Notation

The basic idea in Polish notation is that a binary operator appears just to the left of the two operands rather than in between the two operands. Thus $x + y$ is written as $+ xy$.

The translation of expressions from procedural language into Polish notation is extremely important in compiling and can be accomplished by first representing the given expression by means of an arborescence as follows:

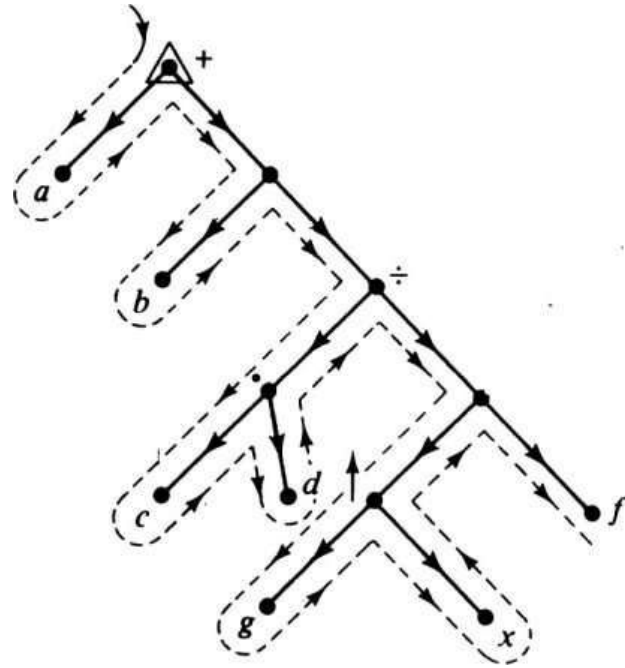
- Each variable (or constant) appearing in the expression is represented by a pendant vertex.
- Each internal vertex represents a binary operator having the two sub-arborescences as its operands.

Application of Arborescence - Polish Notation

To obtain the expression in Polish notation, we traverse the arborescence starting from the root from left to right and from top to bottom, as indicated by the dotted line in this figure.

Each time we come across a vertex that has not been traversed before, we append its label to the existing string. Thus we get the following:

$$+ a - b \div \cdot cd - \uparrow gxf.$$



Application of Arborescence - Polish Notation

An expression in Polish notation is evaluated as follows:

We start at the right extreme and move to the left.

Whenever an operator is encountered the operation is performed between the two operands immediately to the right of it.

After an operation is performed, the resultant is regarded as one operand for the next operation.

The advantage of this method is that there are no parentheses and the operators appear in the order (from right to left) in which they are to be acted upon. Therefore, no back and forth scanning is required.

Ordered Trees

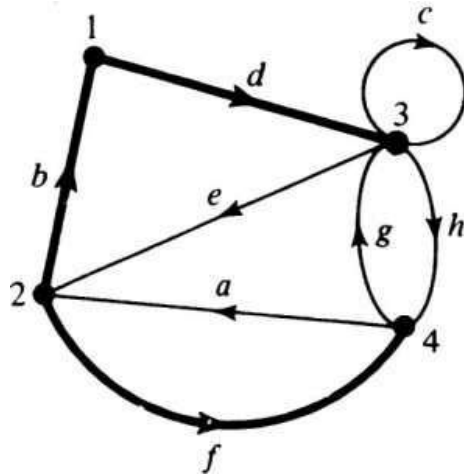
Notice that in the expression arborescence for the Polish Notation conversion, the relative positions of the vertices in the plane of the paper are important and must be preserved.

In this sense it is a “rigid” graph, and a graph isomorphic to it may not preserve its properties. In computer literature a tree in which the relative order of subtrees meeting at each vertex must be preserved is called an **ordered tree** or a **planar tree**.

Spanning Arborescence

A spanning tree in an n -vertex connected digraph, analogous to a spanning tree in an undirected graph, consists of $n - 1$ directed edges.

A spanning arborescence in a connected digraph is a spanning tree that is also an arborescence. For example, $\{f, b, d\}$ is a spanning arborescence of this graph.



Spanning Arborescence and Euler Line

There is a striking relationship between a spanning arborescence and an Euler line, which is shown by **Theorem 9-4** and **9-5**.

THEOREM 9-4: *“In a connected, balanced digraph G of n vertices and m edges, let W be an Euler line, which starts and ends at a vertex v . Among the m edges in W there are $n - 1$ edges that “enter” each of $n - 1$ vertices, other than v , for the first time. The subdigraph g of these $n - 1$ directed edges together with the n vertices is a spanning arborescence of G , rooted at vertex v .”*

In the previous graph, $W = (b \ d \ c \ e \ f \ g \ h \ a)$ is an Euler line, starting and ending at vertex 2. The subdigraph $\{b, d, f\}$ is a spanning arborescence rooted at vertex 2.

Spanning Arborescence and Euler Line

Theorem 9-4 provides a method of obtaining a spanning arborescence rooted at any specified vertex, provided the digraph is Eulerian.

Conversely, given a spanning arborescence in an Euler digraph, an Euler line can be constructed using **Theorem 9-5**. This important result is used in counting the **number of distinct Euler lines**.

THEOREM 9-5

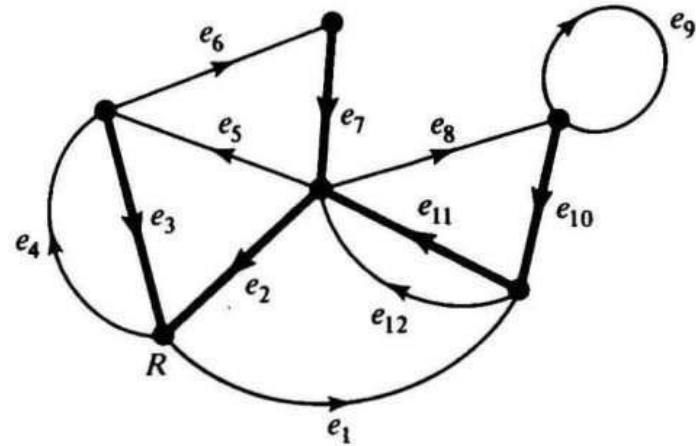
Let G be an Euler digraph and T be a spanning in-tree in G , rooted at a vertex R . Let e_1 be an edge in G incident out of the vertex R . Then a directed walk $W = (e_1, e_2, \dots, e_m)$ is a directed Euler line, if it is constructed as follows:

1. No edge is included in W more than once.
2. In exiting a vertex the one edge belonging to T is not used until all other outgoing edges have been traversed.
3. The walk is terminated only when a vertex is reached from which there is no edge left on which to exit.

THEOREM 9-5

For the sake of traversing the edges along with rather than opposite to the direction of edges, it is better to express **Theorem 9-5** in terms of an **in-tree**, that is, an arborescence in which the direction of every edge has been inverted.

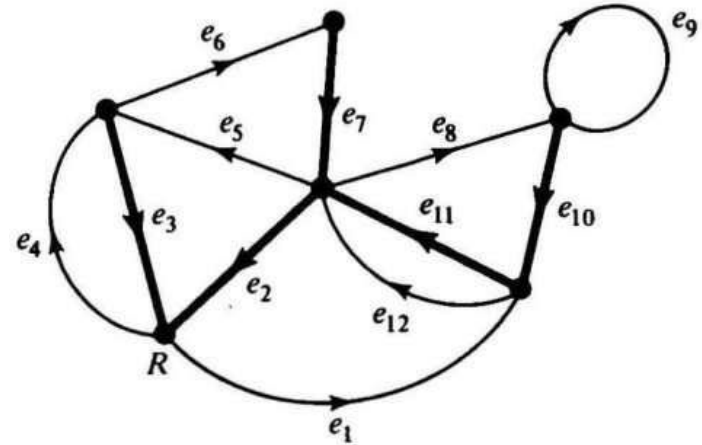
In this figure, the subdigraph $\{e_2, e_3, e_7, e_{10}, e_{11}\}$ is a spanning in-tree.



THEOREM 9-5

The number of distinct Euler lines formed from a given in-tree T , and starting with edge e_1 at R , can be computed by considering all the choices available at each vertex, after starting with e_1 .

Since there is exactly one outgoing edge in T at each vertex and this edge is to be selected last (rule 2), the remaining $d^+(v_i) - 1$ edges at vertex v_i can be chosen in $[d^+(v_i) - 1]!$ ways.



THEOREM 9-5

And since these are independent choices, we have altogether

$$\prod_{i=1}^n [d^+(v_i) - 1]!$$

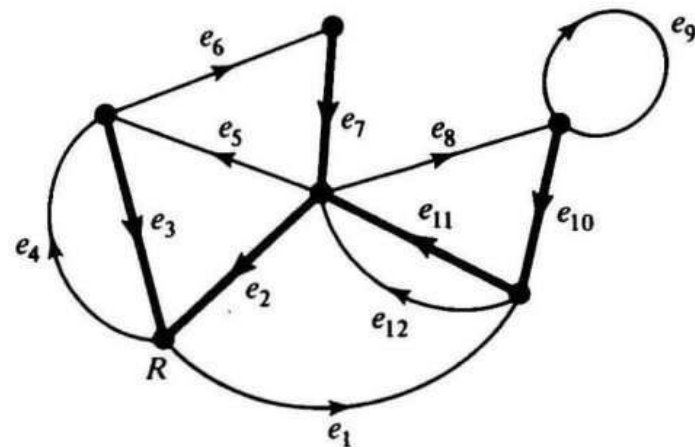
different Euler lines.

Applying these rules in the in-tree $\{e_2, e_3, e_7, e_{10}, e_{11}\}$, starting with edge e_1 . We get the following two directed Euler lines:

$(e_1 e_{12} e_5 e_6 e_7 e_8 e_9 e_{10} e_{11} e_2 e_4 e_3)$, and

$(e_1 e_{12} e_8 e_9 e_{10} e_{11} e_5 e_6 e_7 e_2 e_4 e_3)$.

Note that these are not all the directed Euler lines in the digraph, but only those that are generated by this specific in-tree



FUNDAMENTAL CIRCUITS IN DIGRAPHS

Just as in the case of undirected graphs, every chord c_i when added to the spanning tree T produces a fundamental circuit (which may be a directed circuit or a semicircuit).

A cut-set in a connected digraph G (just as in an undirected graph) induces a partitioning of the vertices of G into two disjoint subsets V_1 and V_2 such that the cut-set consists of all those edges that have one end vertex in V_1 and the other in V_2 .

FUNDAMENTAL CIRCUITS IN DIGRAPHS

Rank $r = 5$

Nullity $\mu = 4$

Spanning tree $T: \{a, d, f, h, k\}$

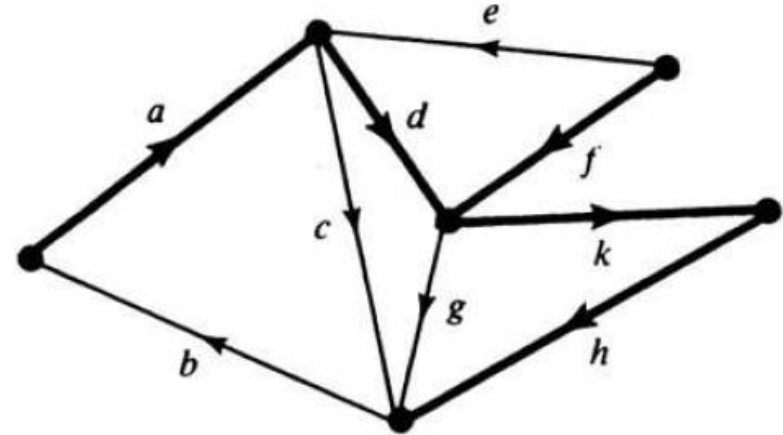
Chord-set with respect
to $T: \{b, c, e, g\}$

Fundamental circuits
with respect to T

- $\left\{ \begin{array}{ll} d f e & \text{(semicircuit)} \\ d k h c & \text{(semicircuit)} \\ k h g & \text{(semicircuit)} \\ a d k h b & \text{(directed circuit)} \end{array} \right.$

Fundamental cut-sets
with respect to T

- $\left\{ \begin{array}{l} a b \\ b c d e \\ e f \\ b c g k \\ b c g h \end{array} \right.$



Set of Directed Circuits

Any circuit (directed or semicircuit in a digraph) can be obtained as a ring sum of some of the fundamental circuits.

But in many applications we are interested only in the set of directed circuits in a digraph. Is there a similar basis for all directed circuits?

The answer, unfortunately, is no. The ring sum of two directed circuits is not necessarily another directed circuit or edge-disjoint union of directed circuits.

For example, in the previous graph, the ring sum of directed circuits $a \ c \ b \oplus a \ d \ g \ b$
 $= d \ g \ c$, a semicircuit.

MATRICES A, B, AND C OF DIGRAPHS

In order to account for the orientation of the edges, the incidence (A), circuits (B), and cut-set (C) matrices consist of $+1$, 0 , -1 (instead of only 0 and 1 for undirected graphs).

The numbers $+1$, 0 , -1 are regarded as ordinary real numbers. Their addition and multiplication are interpreted as in ordinary arithmetic (and not modulo 2 arithmetic as in the case of undirected graphs).

Consequently, the vectors and vector spaces associated with a digraph and its subdigraphs are over the field of all real numbers, and not $\text{GF}(2)$.

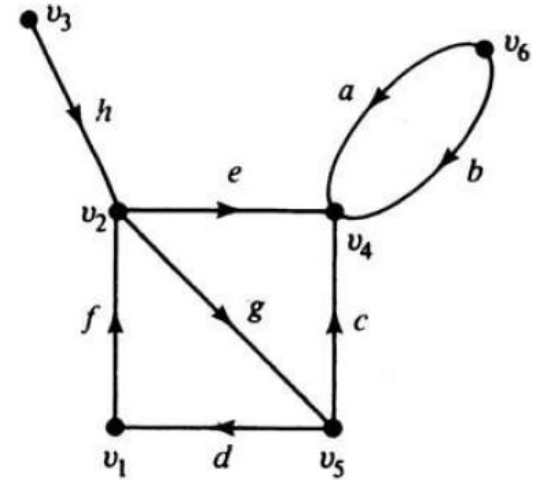
Incidence Matrix

The incidence matrix of a digraph with n vertices, e edges, and no self-loops is an n by e matrix $A = [a_{ij}]$, whose rows correspond to vertices and columns correspond to edges, such that

$$\begin{aligned} a_{ij} &= 1, \text{ if } j^{\text{th}} \text{ edge is incident out of } i^{\text{th}} \text{ vertex,} \\ &= -1, \text{ if } j^{\text{th}} \text{ edge is incident into } i^{\text{th}} \text{ vertex,} \\ &= 0, \text{ if } j^{\text{th}} \text{ edge is not incident on } i^{\text{th}} \text{ vertex.} \end{aligned}$$

Incidence Matrix

$$\begin{array}{c} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ v_6 \end{array} \begin{bmatrix} a & b & c & d & e & f & g & h \\ 0 & 0 & 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ -1 & -1 & -1 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & -1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$



Unimodularity of A

If A is the incidence matrix of an undirected graph, the determinant of every square submatrix of A is either 0 or 1.

In the case of digraphs, the incidence matrix A is in the real field, and on first sight it would appear that the determinants of its square submatrices could acquire any integral value. This, however, is not the case, as shown in the following important theorem.

THEOREM 9-7

“The determinant of every square submatrix of A , the incidence matrix of a digraph, is 1, -1 , or 0.”

Proof: The theorem can be proved directly by expanding the determinant of a square submatrix of A .

Consider a k by k submatrix M of A . If M has any column or row consisting of all zeros, $\det M$ is clearly zero. Also $\det M = 0$ if every column of M contains the two nonzero entries, a 1 and a -1 .

Now if $\det M \neq 0$ (i.e., M is nonsingular), then the sum of entries in each column of M cannot be zero. Therefore, M must have a column in which there is a single nonzero element that is either $+1$ or -1 . Let this single element be in the (i, j) th position in M . Thus, $\det M = \pm 1 \cdot \det M_{ij}$

THEOREM 9-7

The $(k-1)$ by $(k-1)$ submatrix M_{ij} is also nonsingular (because M is nonsingular); therefore, it too must have at least one column with a single nonzero entry, say in the (p, q) th position.

Expanding $\det M_{ij}$ about this element in the (p,q) th position,

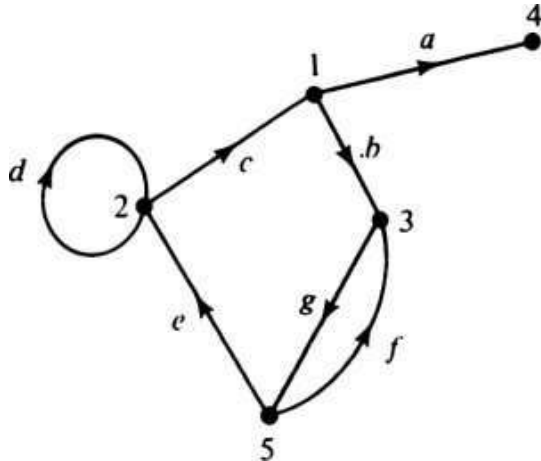
$$\det M_{ij} = \pm [\text{determinant of a nonsingular } (k-2) \text{ by } (k-2) \text{ submatrix of } M]$$

Repeated application of this procedure yields $\det M = \pm 1$.

ADJACENCY MATRIX OF A DIGRAPH

Let G be a digraph with n vertices, containing no parallel edges. Then the adjacency matrix $X = [x_{ij}]$ of the digraph G is an n by n $(0, 1)$ -matrix whose element

$x_{ij} = 1$, if there is an edge directed from i^{th} vertex to j^{th} vertex,
= 0, otherwise



$$X = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} 0 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{bmatrix} \end{matrix}$$

ADJACENCY MATRIX OF A DIGRAPH

The following observations are made on the properties of the adjacency matrix X :

1. X is a symmetric matrix if and only if G is a symmetric digraph.
2. Every nonzero element on the main diagonal represents a self-loop at the corresponding vertex.
3. There is no way of showing parallel edges in X . This is why the adjacency matrix is defined only for a digraph without parallel edges.
4. The sum of each row equals the out-degree of the corresponding vertex, and the sum of each column equals the in-degree of the corresponding vertex.
The number of nonzero entries in X equals the number of edges in G .

ADJACENCY MATRIX OF A DIGRAPH

5. Permutation of any two rows accompanied by a permutation of the corresponding columns does not alter the graph. The permutation merely corresponds to a reordering of the vertices. Thus two digraphs are isomorphic if and only if their adjacency matrices differ only by such permutations.
6. If X is the adjacency matrix of a digraph G , then the transposed matrix X^T is the adjacency matrix of a digraph G^R obtained by reversing the direction of every edge in G .
7. For any square $(0, 1)$ -matrix Q of order n , there exists a unique digraph G of n vertices, such that Q is the adjacency matrix of G .

Connectedness and the Adjacency Matrix

A digraph is disconnected if and only if its vertices can be ordered in such a way that its adjacency matrix X can be expressed as the direct sum of two square submatrices X_1 and X_2 as follows

$$X = \left[\begin{array}{c|c} X_1 & 0 \\ \hline 0 & X_2 \end{array} \right]$$

Such a partitioning is possible if and only if the vertices in the submatrix X_1 have no edge going to or coming from the vertex set in X_2 .

Connectedness and the Adjacency Matrix

Similarly, a digraph is weakly connected if and only if its vertices can be ordered in such a way that its adjacency matrix X can be expressed in either of the following two forms:

$$X = \begin{bmatrix} X_1 & 0 \\ X_{21} & X_2 \end{bmatrix}$$

$$X = \begin{bmatrix} X_1 & X_{12} \\ 0 & X_2 \end{bmatrix}$$

A digraph that is neither disconnected nor weakly connected is strongly connected. Therefore, we conclude that a digraph G is strongly connected if and only if the vertices of G cannot be ordered in any of the previous three forms.

Number of Edge Sequences

THEOREM 9-10: *“The $(i,j)^{th}$ entry in X^r equals the number of different, directed edge sequences of r edges from the i^{th} vertex to the j^{th} .”*

Proof: We prove this by Induction. First, the theorem is trivially true for $r = 1$.

For the inductive hypothesis, assume that the theorem holds for X^{r-1} . Now, for expanding to the $(i, j)^{th}$ entry in $X^r (= X^{r-1} \cdot X)$

$$\begin{aligned} &= \sum_{k=1}^n [(i, k)^{th} \text{ entry in } X^{r-1}] \cdot x_{kj} \\ &= \sum_{k=1}^n (\text{number of all directed edge sequences} \\ &\quad \text{of length } r - 1 \text{ from vertex } i \text{ to } k) \cdot x_{kj}, \end{aligned}$$

Number of Edge Sequences

Here, $x_{kj} = 1$ or 0 , depending on whether or not there is a directed edge from k to j . Thus a term in the sum is nonzero if and only if there is a directed edge sequence of length r from i to j , whose last edge is from k to j . If the term is nonzero, its value equals the number of such edge sequences from i to j via k . This holds for every vertex k , $1 \leq k \leq n$.

Therefore, the X^r is equal to the number of all possible directed edge sequences from i to j .

Number of Edge Sequences

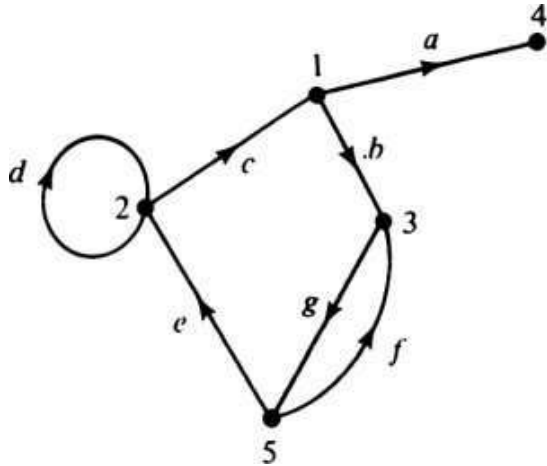
X^r gives us the number of all possible edge sequences which can fall in three different categories:

1. Directed paths from i to j : Those directed edge sequences in which no vertex is traversed more than once.
2. Directed walks from i to j : Those directed edge sequences in which a vertex may be traversed more than once, but no edge is traversed more than once.
3. Those directed edge sequences in which an edge may also be traversed more than once.

Unfortunately, there is no easy way of separating these.

Number of Edge Sequences

For example, let us examine the fourth power of the adjacency matrix of the following graph



$$X = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} 0 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{bmatrix} \end{matrix} \quad X^4 = \begin{bmatrix} 1 & 1 & 0 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 \\ 1 & 2 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 2 & 2 & 1 & 1 & 2 \end{bmatrix}.$$

The entry in the second row and third column represents two directed edge sequences of length four: c b g f (a directed walk from 2 to 3) and d d c b (not a walk).

Number of Arborescences

Counting of the spanning arborescences in any connected digraph is a trivial extension of counting them in a simple, connected digraph.

The self-loops can be discarded right away, and addition of a parallel edge **b** to an existing edge **a** simply doubles the number of the arborescences, by repeating the same arborescences with **a** replaced by **b**.

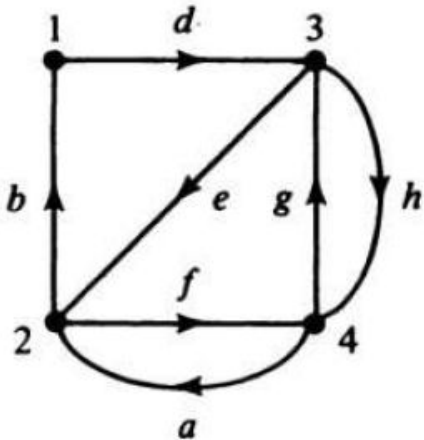
So, when we apply the arborescence counting formula, we will at first convert any connected digraph into a simple connected digraph.

Number of Arborescences

In preparation for the arborescence counting formula, let us define, for a simple digraph G of n vertices, an n by n matrix called the Kirchhoff matrix $K(G)$:

$k_{ii} = d^-(v_i)$, in-degree of the i^{th} vertex,

$k_{ij} = -x_{ij}$, $(i, j)^{\text{th}}$ entry in the adjacency matrix, with a negative sign



$$K(G) = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 1 & 0 & -1 & 0 \\ -1 & 2 & 0 & -1 \\ 0 & -1 & 2 & -1 \\ 0 & -1 & -1 & 2 \end{bmatrix} \end{matrix}$$

Number of Arborescences

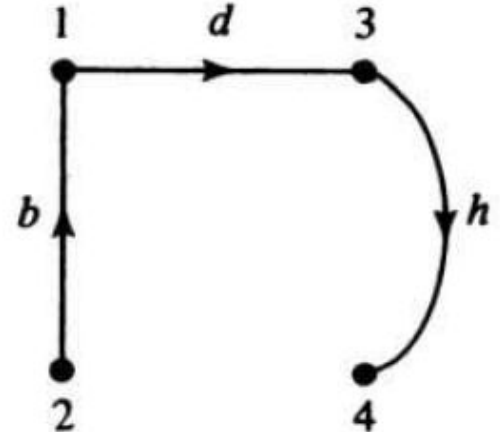
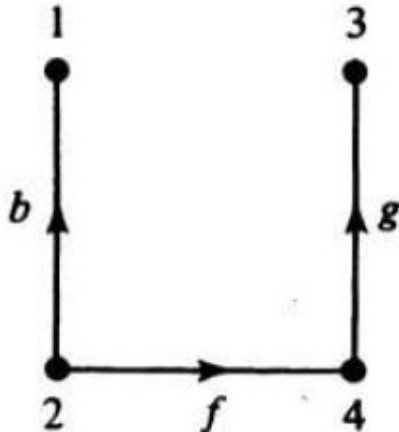
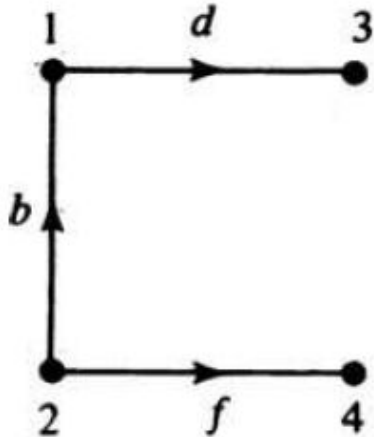
THEOREM 9-11: *“A simple digraph G of n vertices and $n - 1$ directed edges is an arborescence rooted at v_1 if and only if the $(1, 1)$ cofactor of $K(G)$ is equal to 1.”*

THEOREM 9-12: *“Let $K(G)$ be the Kirchhoff matrix of a simple digraph G . Then the value of the (q, q) cofactor of $K(G)$ is equal to the number of arborescences in G rooted at the vertex v_q .”*

Number of Arborescences

Applying Theorem 9-12 for the previous graph, we see that the cofactor of every entry in the second row of the K matrix is 3.

The digraph does indeed have three arborescences rooted at vertex 2.



Number of Distinct Euler Lines

In fact, for an Euler digraph G , all cofactors of $K(G)$ are equal, because the sum of each row and the sum of each column equals zero. Let this common value of all cofactors of $K(G)$ be σ .

This σ is the number of different arborescences rooted at any given vertex in G .

THEOREM 9-13: *In an Euler digraph the number of Euler lines is*

$$\sigma \cdot \prod_{i=1}^n [d^-(v_i) - 1]!.$$

Number of Distinct Euler Lines

Recall the graph we encountered in the Teleprinter's problem. Its K matrix is

$$\begin{bmatrix} 1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & -1 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 2 & -1 & 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & -1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 2 & -1 \\ -1 & -1 & 0 & 0 & 0 & 0 & 0 & 2 \end{bmatrix}.$$

The cofactor of any term in this matrix is 16, and therefore $\sigma = 16$ in Theorem 9-13. Since $d^-(v_i) = 2$ for each v_i , so the product of all the terms becomes 1.

Therefore, the number of Euler lines is 16.

PAIRED COMPARISONS AND TOURNAMENTS

In many experiments, one is required to rank a number of given objects by comparing only two at a time. This is called the method of paired comparisons.

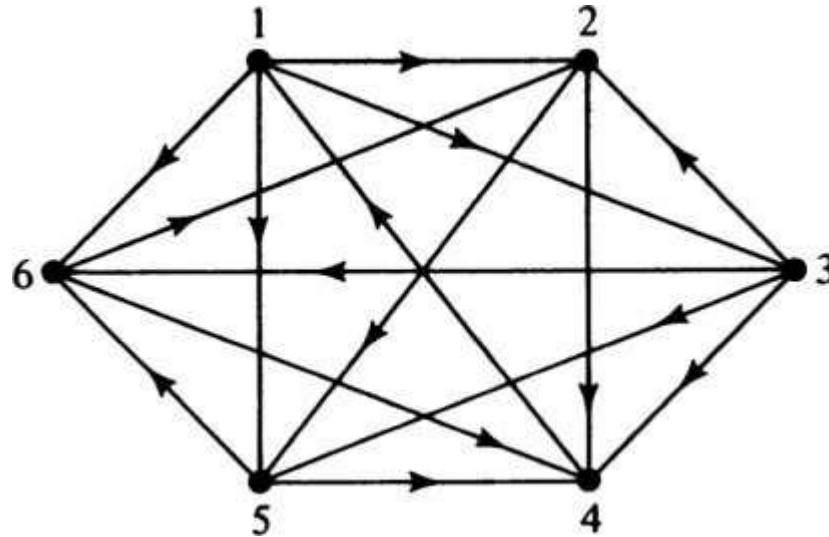
A digraph is a natural way of representing the results of a paired-comparison experiment.

Suppose six different dog foods $\{1, 2, \dots, 6\}$ are to be ranked. Each day two of the six delicacies are served to a dog, and the dog established preference for one food over the other according to which plate he finished first.

The experiment is conducted for 15 days, so that all possible pairs could be tried.

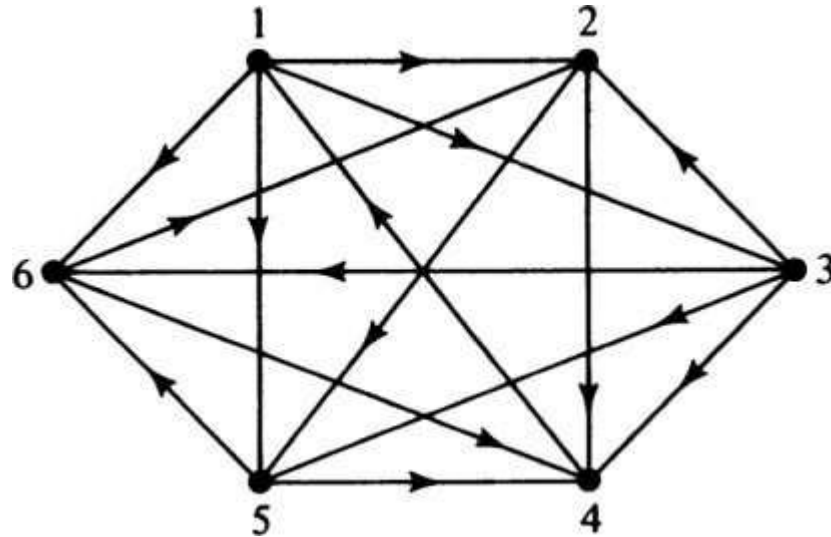
PAIRED COMPARISONS AND TOURNAMENTS

In the graph representation, an edge is drawn from the preferred dish to the less preferred. For example, 1 was preferred to 2 in the graph below. Such a graph is called a preference graph.



PAIRED COMPARISONS AND TOURNAMENTS

Establishing a rank from a given preference graph is, in general, not easy. In this graph, for example, due to some canine inconsistency, the dog preferred food 1 over 2, 2 over 4, and then 4 over 1. So which of the three is the best?



PAIRED COMPARISONS AND TOURNAMENTS

A similar situation is encountered in tournaments.

The results of a round-robin tournament in which every player has played against every other may also be represented by a digraph in which an edge directed from vertex a to b represents the victory of player a over player b . This is why a complete asymmetric digraph was called a **tournament**.

Ranking by Score

A straightforward method of ranking, and the one that has been traditionally used in round-robin tournaments, is to rank each player by his **score**.

The score is the number of games the player has won. In terms of the dog food, the number of times the particular dish was preferred is its score.

The score of a player in a tournament equals the out-degree of the corresponding vertex in the digraph. Thus if we use the scores for ranking, we would rank the six dog foods as (1,3), (2,5,6), and 4.

Ranking by Hamiltonian Path

Ranking the vertices according to their out-degrees is not always a satisfactory method, although it is the easiest. In particular, this method loses significance if the tournament is incomplete (that is, the players do not compete in the same number of games).

Another method sometimes used is to rank the players in a **directed Hamiltonian path**, such that each player has defeated his successor. One such ranking in the previous graph is 1 3 2 5 6 4.

THEOREM 9-14

“Every complete tournament has a directed Hamiltonian path.”

Proof: The theorem will be proved by induction on the number of vertices. By actual sketching, the theorem can be shown to hold for all complete tournaments of 1, 2, 3, and 4 vertices.

Let us make the inductive assumption that the theorem is true for all complete tournaments of n vertices, and then prove that it also holds for all tournaments of $n + 1$ vertices.

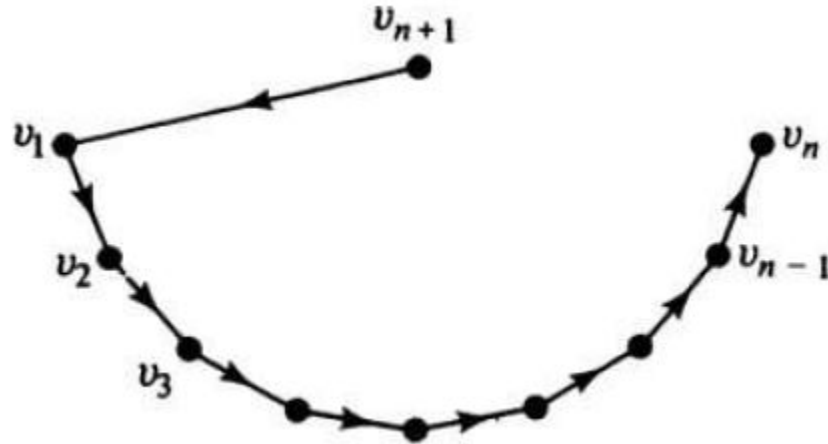
THEOREM 9-14

Let G be any complete tournament of $n + 1$ vertices. Let g be an n -vertex complete subtournament of G . By inductive assumption, g has a directed Hamiltonian path. Let that path be $v_1 v_2 \dots v_n$.

Let the vertex present in G but not in g be called v_{n+1} . Since G is a complete tournament of $n + 1$ vertices, the vertex v_{n+1} in G has a directed edge either to or from each of the other vertices $v_1 v_2 \dots, v_n$. The following three cases are possible.

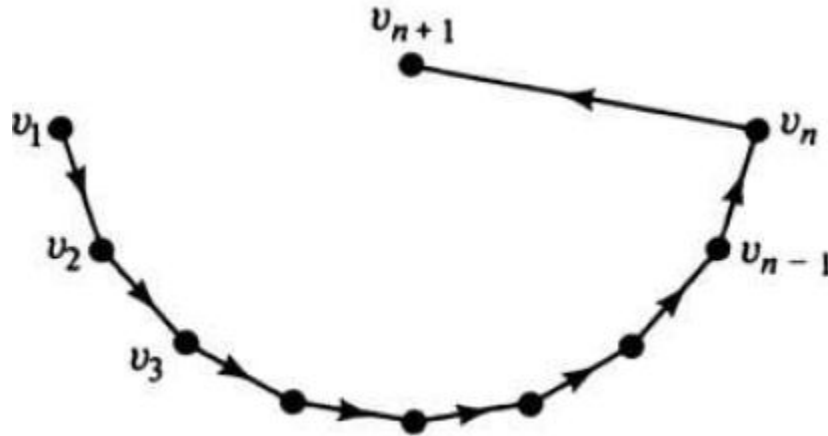
THEOREM 9-14

Case 1: The edge between v_{n+1} and v_1 is directed toward v_1 . Then we have a Hamiltonian path $v_{n+1} v_1 v_2 \dots v_n$ in G , and the proof is complete.



THEOREM 9-14

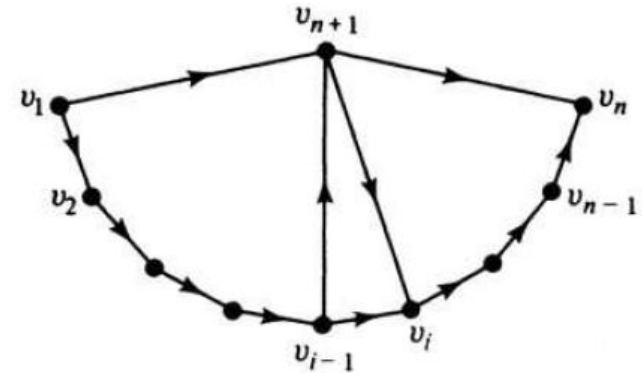
Case 2: There is an edge, directed from v_n to v_{n+1} . Then also we have a Hamiltonian path in G , which is $v_1 v_2 \cdots v_n v_{n+1}$ and the proof is complete.



THEOREM 9-14

Case 3: Instead, both these edges are directed from v_1 to v_{n+1} and from v_{n+1} to v_n . In this case, as we move from v_1 to v_n , we encounter a reversal of direction in the edges incident on v_{n+1} . This reversal must occur because edge (v_1, v_{n+1}) is directed toward v_{n+1} but edge (v_n, v_{n+1}) is directed away from v_{n+1} .

Call the vertex at which the first such reversal occurs v_i (v_i may be v_n itself). Then edge (v_{i-1}, v_{n+1}) must be directed toward v_{n+1} . In this case we have a directed Hamiltonian path $v_1 v_2 \dots v_{i-1} v_{n+1} v_i v_{i+1} \dots v_n$ in G .



THEOREM 9-14

However, this method of ranking also suffers from some drawbacks.

For one, there may be discrepancies between such a ranking and the scores of the players.

Second, a tournament may have more than one directed Hamiltonian path, and therefore several different rankings are possible.

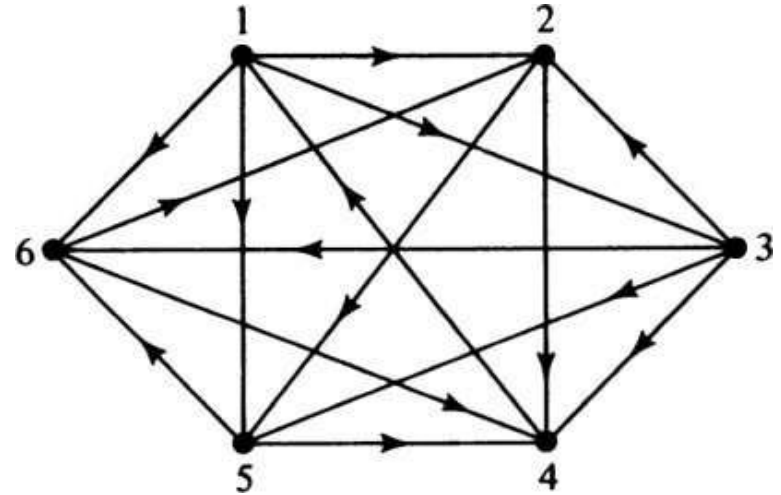
For instance, in the dog food preference graph, 1 3 2 5 6 4 and 1 3 5 6 2 4 are two different Hamiltonian rankings.

Ranking with Minimum Violations

For a given ranking of the n vertices in any tournament (complete or incomplete), a violation is defined as an edge directed from v_i to v_j if v_j precedes v_i in the ranking.

For example, in the dog food preference graph, the order 1 3 2 5 6 4 has the following two violations: edges 4 to 1 and 6 to 2.

The order 3 2 5 6 4 1 has five violations, edges 1 to 3, 1 to 2, 6 to 2, 1 to 5, and 1 to 6.



Ranking with Minimum Violations

Ranking with the minimum number of violations represents the fewest possible upsets for a given tournament. It can be shown that a ranking with minimum violations automatically includes the ranking according to scores, as well as a Hamiltonian ranking.

Moreover, a minimum-violation ranking is also meaningful for incomplete tournaments. Thus this may be considered the best method of ranking.

However, out of all $n!$ possible orders of n vertices, to find one with minimum violations is computationally difficult. A method using dynamic programming has been used and is the best available so far, but it is computationally slow and cumbersome.